DS 6021: Intro to Predictive Modeling
Lab 1: Statistical Inference, Sample Comparisons, A/B Testing

September 4, 2026

Due September 6, 2026 11:59pm

Materials Required

All available in our class github repository:

1. Penguins.csv – Palmer penguins dataset
2. DS6021-Fall26-Anonymized.csv – In class dataset
3. Clinical_trial.csv – New dataset that has been uploaded to our class repository
4. In-class notebooks from 8/31 (comparing_samples.ipynb) and 9/2 (statistical_power.ipynb)

Note: Many of the questions are open ended and allow for multiple perspectives and correct answers.

Activity 1: Comparing two proportions

Using the Palmer penguins dataset, we want to explore comparing two proportions, which we did not get to in class. We answer the following question: Are penguins that live on Biscoe island larger? We want to look at the "rate" of large penguins on Biscoe vs the other two islands. Please use the following definitions and resources:

- What does larger mean? For any penguin, is it larger than median penguin body mass. We can write this as: `peng["is_large"] = peng["body_mass_g"] > peng["body_mass_g"].median()`
- You are going to want to look at the fraction of large penguins on Biscoe island vs large penguins on other islands
- Stats models has a proportions_ztest function that will be useful for this exercise
- The pandas crosstab function might also be useful

Please complete the following:

1. [2pts] Visualize body mass of the penguins across different islands. What does this tell you?
2. [3pts] Using the definitions above, answer the question: Are penguins on Biscoe island larger?
3. [2pts] Now visualize which species of penguins live on each island. How does that change your answer to the question: Are penguins on Biscoe island larger?
4. [3pts] Repeat the proportion test considering only one species of penguin. Which species did you choose, and how did that impact your answer?

Activity 2: Digging Further into ANOVA with class data

In class, we looked a little bit at our class data to explore whether and how average number of hours of sleep per night as a function of years since undergrad.

1. [2 points] Re-bin the data into three groups instead of 5: [<1 year | 1-3 years | 4+ years] and plot hours of sleep for these three groups.
2. [2 points] Repeat the test for equal variance with these three groups, as well as the ANOVA. What conclusions can you draw? Are any other assumptions violated?

3. [4 points] Reflection: One thing we have been ignoring in this analysis is the fact that these are ordered variables: e.g. they aren't independent buckets like in our penguins dataset where the different species have no inherent ordering. Here, <1 year means something in relation to 1-3 years and so forth. Please answer the following questions:
    a. How are the original buckets reflective of a poor survey design?
    b. Does splitting our data into three buckets vs five improve the survey design?
    c. If we were redesigning this survey, what would you recommend in order to answer the question about whether or not students who graduated more recently get more or less sleep than students who graduated longer ago?
4. [2 points] Reflection: We've been using the in-class survey data to help understand concepts around statistical inference, sample comparisons, and data cleaning. Is this a good dataset to help understand these concepts? Please explain why or why not, and reflect on what this might tell you more broadly about data collection, survey design, and drawing conclusions from a particular set of data.

Activity 3: Migraine Drug Formulations

A pharmaceutical company tested three formulations of a pain relief medicine for migraine headache sufferers. 27 volunteers were selected and 9 were randomly assigned to one of the three drug formulations. Subjects took the drug during their next migraine episode and reported pain on a 1–10 scale (1 = no pain, 10 = extreme pain) 30 minutes after taking the drug. The dataset, `Clinical_trial`, contains the results of this experiment.

Note: Some of the questions are purposefully left open-ended as there are multiple approaches to solve these problems.

1. [2pts] Visualize the pain ratings across the three drug formulations. Provide a brief interpretation of what the graph reveals.
2. [5pts] As the Data Scientist on a team of researchers at the pharmaceutical company, your main task is to evaluate which drug formulation most effectively reduces migraine pain. Formulate the appropriate hypotheses, perform the relevant statistical test(s), and communicate each step of your analysis and results in clear, accessible written language for non-technical team members.
3. [3pts] Suppose we wanted to build a supervised learning model using this dataset. What would be the prediction goal? Note that you do not have to build a predictive model to answer this question. Discuss in what situations a pharmaceutical company might use statistical inference vs prediction.

Activity 4: Statistical Power

In class, we did some simulations to explore concepts around statistical power. We specifically looked at an example of an A/B test to understand how long it would take us to detect a 5% difference in average sales when comparing two websites. Now let's assume you are presented with the following scenario:

A business analyst (BA) comes to you needing help making a decision. So far, they have been running an A/B test for 4 days between two versions of an app, and they want to look at the mean length of time spent on the app, with more time being better. They have presented you with the following information:

- Each day, 300 users are in the control (A) and treatment (B) groups.
- A is the version of the app currently in production, and users typically spend an average of 12 minutes on the App with a standard deviation of +/- 4 minutes.
- After running this A/B test for 4 days, they have observed that users who receive version B of the app spend an average of 12.5 minutes on the App vs 12 for version A.

This BA is getting some pressure from their leadership to move everyone to version B of the app, but they want you to tell them whether or not they can trust the results they have so far.

1. [3pts] Assuming an alpha of 0.05 and a power of 0.8, what is the minimum effect size we could measure after 4 days?
2. [3pts] Based on your answer to the first question, would you recommend discontinuing the A/B test and moving everyone to the B version of the app? If yes, why? If not, why not?
3. [5pts] Create a visualization and three bullet points that you could present to leadership about your analysis to help them make a decision.
4. [4pts] What if your BA collaborator came back to you and said the test was being discontinued (regardless of your analysis)—how would that change how you frame the results to leadership?

Deliverables

For each activity, please include the following:

- All code used to complete each of the activities. This can be turned in as a single Jupyter notebook, multiple Jupyter notebooks, or one or more .py files. Up to you!
- For open ended questions, please include a few sentences or a paragraph answering the question. This text can be included in your Jupyter notebook as a markdown cell or a separate document.
- Per our policy on Generative AI detailed in the syllabus, if you use it to generate code, you need to flag how and where you used it, and how you ascertained the correctness of what you did. Note: You are not allowed to use Generative AI for any of the reflection questions.